Course Mechanics

- Syllabus has all the class info and is available in BYS
- Course requirements
  - Midterm Exam (date TBD): 60%
  - Final Exam (date TBD): 40%

Major topics & outline

- Fundamental Dimensions and Units
- Introduction to Linear Algebra
  - Vectors, matrices, row-reduced forms
  - Solutions of linear equations, matrix inverse, eigenvalues and eigenvectors
- Introduction to Electrical Circuits
  - Kirchhoff's Current Law
  - Kirchhoff's Voltage Law
- Introduction to Logic Circuits
  - Combinatorial Logic Circuits
  - Implementation of Logic Expressions
- Introduction to Electronics
  - Operational Amplifiers
  - Diodes
  - Capacitors
  - Inductors

Fundamental Dimensions and Units

- The physical quantities used to fully describe natural events and our surroundings are:
  - Length
  - Mass
  - Time
  - Temperature
  - Electric current
  - Amount of substance

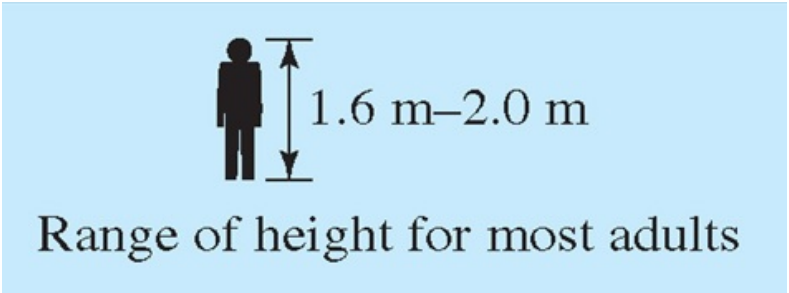
Units

- Used to measure physical dimensions
- Appropriate divisions of physical dimensions to keep numbers manageable
  - 19 years old instead of 612,000,000 seconds old
- Common systems of units
  - International System (SI) of Units
  - British Gravitational (BG) System of Units
  - U.S. Customary Units

Units – SI

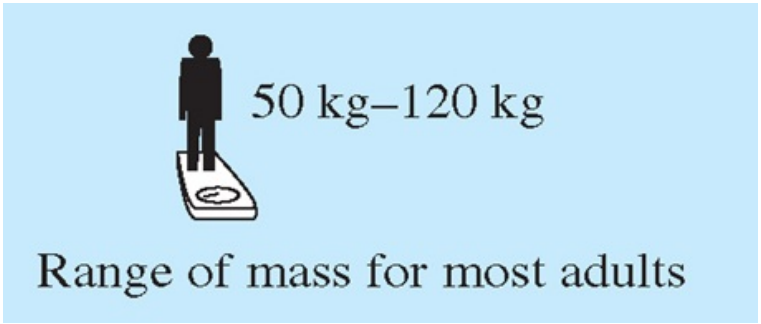
- Most common system of units used in the world
- Examples of SI units are: kg, N, m, cm
- Approved by the General Conference on Weights and Measures (CGPM)
- Series of prefixes & symbols of decimal multiples (adapted by CGPM, 1960)

Fundamental Unit of Length



- **meter (m)** – length of the path traveled by light in a vacuum during a time interval of 1/299,792,458 of a second

Fundamental Unit of Mass



- **kilogram (kg)** – a unit of mass; it is equal to the mass of the international prototype of the kilogram

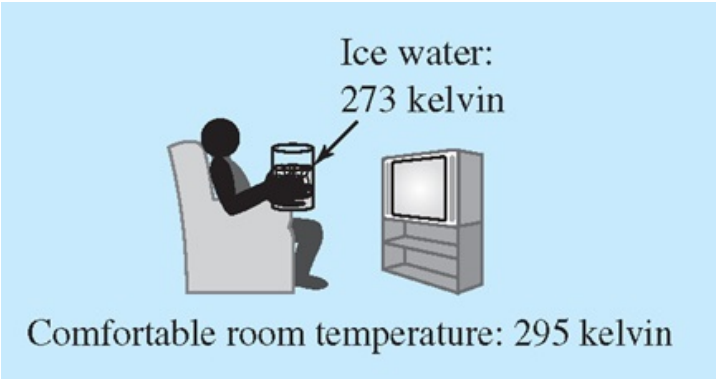
Fundamental Unit of Time



Fastest person can run
100 meters in approximately
10 seconds

- **second (s)** – duration of 9,192,631,770 periods of the radiation corresponding to the transition between the 2 hyperfine levels of the ground state of cesium 133 atom

Fundamental Unit of Temperature



- **kelvin (K)** – unit of thermodynamic temperature, is the fraction 1/273.16 of thermodynamic temperature of the triple point of water

Fundamental Unit of Amount of Substance

Uranium	238	← One of the
Gold	197	heaviest
Silver	108	atoms known
Copper	64	
Calcium	40	
Aluminum	27	
Carbon	12	← Common Carbon is
Helium	4	used as a standard
Hydrogen	1	← Lightest atom

- **mole (mol)** – the amount of substance of a system that contains as many elementary entities as there are atoms in 0.012 kilogram of carbon 12

SI – Prefix & Symbol

Multiplication Factors	Prefix	SI Symbol
$1{,}000{,}000{,}000{,}000{,}000{,}000{,}000 = 10^{24}$	yotta	Y
$1{,}000{,}000{,}000{,}000{,}000{,}000 = 10^{21}$	zetta	Z
$1{,}000{,}000{,}000{,}000{,}000 = 10^{18}$	exa	E
$1{,}000{,}000{,}000{,}000 = 10^{15}$	peta	P
$1{,}000{,}000{,}000{,}000 = 10^{12}$	tera	T
$1{,}000{,}000{,}000 = 10^{9}$	giga	G
$1{,}000{,}000 = 10^{6}$	mega	M
$1000 = 10^{3}$	kilo	k
$100 = 10^{2}$	hecto	h
$10 = 10^{1}$	deka	da
$0.1 = 10^{-1}$	deci	d
$0.01 = 10^{-2}$	centi	c
$0.001 = 10^{-3}$	milli	m
$0.000{,}001 = 10^{-6}$	micro	μ
$0.000{,}000{,}001 = 10^{-9}$	nano	n
$0.000{,}000{,}000{,}001 = 10^{-12}$	pico	p
$0.000{,}000{,}000{,}000{,}001 = 10^{-15}$	femto	f
$0.000{,}000{,}000{,}000{,}000{,}001 = 10^{-18}$	atto	a
$0.000{,}000{,}000{,}000{,}000{,}000{,}001 = 10^{-21}$	zepto	z
$0.000{,}000{,}000{,}000{,}000{,}000{,}000{,}001 = 10^{-24}$	yocto	y

Examples of Derived Units in Engineering

Physical Quantity	Name of SI Unit	Symbol for SI Unit	Expression in Terms of Base Units
Acceleration			m/s^2
Angle	Radian	rad	
Angular acceleration			rad/s^2
Angular velocity			rad/s
Area			m^2
Density			kg/m^3
Energy, work, heat	Joule	J	$N \cdot m$ or $kg \cdot m^2/s^2$
Force	Newton	N	$kg \cdot m/s^2$
Frequency	Hertz	Hz	s^{-1}
Impulse			$N \cdot S$ or $kg \cdot m/s$
Moment or torque			$N \cdot m$ or $kg \cdot m^2/s^2$
Momentum			$kg \cdot m/s$
Power	Watt	W	J/s or $N \cdot m/s$ or $kg \cdot m^2/s^3$
Pressure, stress	Pascal	Pa	N/m^2 or $kg/m \cdot s^2$
Velocity			m/s
Volume			m^3
Electric charge	Coulomb	C	$A \cdot s$
Electric potential	Volt	V	J/C or $m^2 \cdot kg/(s^3 \cdot A^2)$
Electric resistance	Ohm	Ω	V/A or $m^2 \cdot kg/(s^3 \cdot A^2)$
Electric conductance	Siemens	S	$1/\Omega$ or $s^3 \cdot A^2/(m^2 \cdot kg)$
Electric capacitance	Farad	F	C/V or $s^4 \cdot A^2/(m^2 \cdot kg)$
Magnetic flux density	Tesla	T	$V \cdot s/m^2$ or $kg/(s^2 \cdot A)$
Magnetic flux	Weber	Wb	$V \cdot s$ or $m^2 \cdot kg/(s^2 \cdot A)$
Inductance	Henry	H	$V \cdot s/A$
Absorbed dose of radiation	Gray	Gy	J/kg or m^2/s^2

British Gravitation (BG) System

- Primary units are
  - foot (ft) for length (1 ft = 0.3048 m)
  - second for time
  - pound (lb) for force (1 lb = 4.448 N)
  - Fahrenheit ($^\circ$F) for temperature

$$T(^\circ F) = \frac{9}{5}T(^\circ C) + 32 \quad T(^\circ R) = \frac{9}{5}T(K)$$

- Slug is unit of mass which is derived from Newton's second law
  - 1 lb = (1 slug)(1 ft/s^2)

U.S. Customary System of Units

- Primary units are
  - foot (ft) for length (1 ft $=$ 0.3048 m)
  - second for time
  - pound mass (lb_m) for mass (1 lb_m $=$ 0.453592 kg, 1 slug $=$ 32.2 lb_m)
- Pound force (lb_f) is defined as the weight of an object having a mass of 1 lb_m at sea level and at a latitude of $45°$, where acceleration due to gravity is 32.2 ft/s^2 (1 lb_f $=$ 4.448 N)

Examples of SI Units in Everyday Use

Examples of SI Unit Usage	SI Units Used
Camera film	35 mm
Medication dose such as pills	100 mg, 250 mg, or 500 mg
Sports:	
swimming	100 m breaststroke or butterfly stroke
running	100 m, 200 m, 400 m, 5000 m, and so on
Automobile engine capacity	2.2 L (liter), 3.8 L, and so on
Light bulbs	60 W, 100 W, or 150 W
Electric consumption	kWh (killo-Watt-hour)
Radio broadcasting signal frequencies	88–108 MHz (FM broadcast band) 0.54–1.6 MHz (AM broadcast band)
Police, fire	153–159 MHz
Global positioning system signals	1575.42 MHz and 1227.60 MHz

Examples of U.S. Customary Units in Everyday Use

Examples of U.S. Customary Unit Usage	U.S. Customary Units Used
Fuel tank capacity	20 gallons or $2.67\,\text{ft}^3$ ($1\,\text{ft}^3 = 7.48$ gallons)
Sports (length of a football field)	100 yd or 300 ft
Power capacity of an automobile	150 hp or 82500 lb·ft/s (1 hp = 550 lb·ft/s)
Distance between two cities	100 miles (1 mile = 5280 ft)

Systems of Units and Conversion Factors

	System of Units			
Dimension	**SI**	**BG**	**U.S. Customary**	**Conversion Factors**
Length	Meter (m)	Foot (ft)	Foot (ft)	1 ft = 0.3048 m 1 m = 3.2808 ft
Time	Second (s)	Second (s)	Second (s)	
Mass	Kilogram (kg)	Slugs*	Pound mass (lb_m)	1 lb_m = 0.4536 kg 1 kg = 2.2046 lb_m 1 slug = 32.2 lb_m
Force	Newton (N) $1N = (1\,kg)\left(1\frac{m}{s^2}\right)$	1 lb_f** = (1 slug)$\left(1\frac{ft}{s^2}\right)$	One pound mass*** weighs one pound force at sea level	1 N = 224.809E-3lb_f 1 lb_f = 4.448N
Temperature	Degree Celsius (°C) or Kelvin (K)**** K = °C + 273.15	Degree Fahrenheit (°F) or degree Rankine (°R) °R = °F + 459.67	Degree Fahrenheit (°F) or degree Rankine (°R) °R = °F + 459.67	°C = $\frac{5}{9}$[°F − 32] °F = $\frac{9}{5}$°C + 32 K = $\frac{5}{9}$°R °R = $\frac{9}{5}$K
Work, Energy	Joule (J) = (1N)(1m)	$lb_f \cdot ft$ = (1 lb_f)(1 ft) Commonly written as ft · lb_f	$lb_f \cdot ft$ = (1 lb_f)(1 ft)	1 J = 0.7375 ft · lb_f 1 ft · lb_f = 1.3558 J 1 Btu = 778.17 ft · lb_f
Power	Watt(W) = $\frac{1\,\text{Joule}}{1\,\text{second}}$ kW = 1000 W	$\frac{lbf \cdot ft}{\text{second}} = \frac{(1lbf)(1\,ft)}{1\,\text{second}}$	$\frac{lbf \cdot ft}{\text{second}} = \frac{(1lbf)(1\,ft)}{1\,\text{second}}$	1 W = 0.7375 $\frac{ft \cdot lb_f}{s}$ 1 hp = 550 $\frac{ft \cdot lb_f}{s}$ 1 hp = 0.7457 kW

*Derived or secondary dimension

**Fundamental dimension

***Note unlike SI and BG systems, the relationship between pound force and pound mass is not defined using Newton's second law

****Note a Temperature value expressed in K reads Kelvin not degree Kelvin

Unit Conversion

- In engineering analysis and design, there may be a need to convert from one system of units to another
- When communicate with engineers in different countries
- Important to learn to convert information from one system of units to another correctly
- **Always show the appropriate units that go with your calculations**
- See the conversion tables given in the book

Example: Unit Conversion

Example

Given a person who is 6' 1" tall and weighs 185 pound force (lb_f), find height and weight in SI units.

Example: Unit Conversion

Example

Given a person who is 6' 1" tall and weighs 185 pound force (lb_f), find height and weight in SI units.

Solution

Person's height, H

$$H = \left(6ft + 1in\left(\frac{1ft}{12in}\right)\right)\left(\frac{0.3048m}{1ft}\right) = 1.854m$$

Person's weight, W

$$W = 185lb_f\left(\frac{4.448N}{1lb_f}\right) = 822.88N$$

Example: Unit Conversion

Example

Given a person is driving a car at a speed of 65 miles per hour (mph) over a distance of 25 miles, find speed and distance in SI units.

Example: Unit Conversion

Example

Given a person is driving a car at a speed of 65 miles per hour (mph) over a distance of 25 miles, find speed and distance in SI units.

Solution

Speed of car, S

$$S = \left(65\frac{miles}{h}\right)\left(\frac{5280ft}{1mile}\right)\left(\frac{0.3048m}{1ft}\right) = 104,607m/h = 104.607km/h$$

or

$$S = \left(104,607\frac{m}{h}\right)\left(\frac{1h}{3600s}\right) = 29.057m/s$$

Distance traveled, D

$$D = 25miles\left(\frac{5280ft}{1mile}\right)\left(\frac{0.3048m}{1ft}\right)\left(\frac{1km}{1000m}\right) = 40.233km$$

Example: Unit Conversion

Example

Given outside temperature is 80°F and has a density of 0.0735 pound mass per cubic foot (lb_m/ft^3), find temperature and density in SI units.

Example: Unit Conversion

Example

Given outside temperature is 80°F and has a density of 0.0735 pound mass per cubic foot (lb_m/ft^3), find temperature and density in SI units.

Solution

Temperature of air, T

$$T(^*C) = \frac{5}{9}\left(T(^\circ F) - 32\right) = \frac{5}{9}(80 - 32) = 26.7^\circ C$$

Density of air, ρ

$$\rho = \left(0.0735\frac{lb_m}{ft^3}\right)\left(\frac{0.453kg}{1lb_m}\right)\left(\frac{1ft}{0.3048m}\right)^3 = 1.176kg/m^3$$

Example: Unit Conversion

Example

Given area $A = 100cm^2$, find A in m^2.

Example: Unit Conversion

Example

Given area $A = 100cm^2$, find A in m^2.

Solution

$$A = 100cm^2 \left(\frac{1m}{100cm} \right)^2 = 0.01m^2$$

Example: Unit Conversion

Example

Given volume $V = 1000mm^3$, find V in m^3.

Example: Unit Conversion

Example

Given volume $V = 1000mm^3$, find V in m^3.

Solution

$$V = 1000mm^3 \left(\frac{1m}{1000mm}\right)^3 = 10^{-6}m^3$$

Example: Unit Conversion

Example

Given atmosphere pressure $P = 10^5 N/m^2$, find P in lb/in^2.

Example: Unit Conversion

Example

Given atmosphere pressure $P = 10^5 N/m^2$, find P in lb/in^2.

Solution

$$P = \left(10^5 \frac{N}{m^2}\right)\left(\frac{1lb}{4.448N}\right)\left(\frac{0.0254m}{1in}\right)^2 = 14.5lb/in^2$$

Example: Unit Conversion

Example

Given density of water $\rho = 1000 kg/m^3$, find ρ in lb_m/ft^3.

Example: Unit Conversion

Example

Given density of water $\rho = 1000 kg/m^3$, find ρ in lb_m/ft^3.

Solution

$$\rho = \left(1000 \frac{kg}{m^3}\right)\left(\frac{1 lb_m}{0.4536 kg}\right)\left(\frac{1 m}{3.28 ft}\right)^3 = 62.5 lb_m/ft^3$$

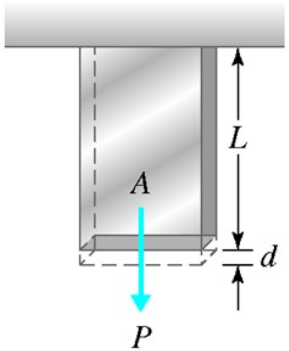
Examples of Unit Conversion

Quantity	SI → U.S. Customary	U.S. Customary → SI
Length	1 mm = 0.03937 in. 1 mm = 0.00328 ft 1 cm = 0.39370 in. 1 cm = 0.0328 ft 1 m = 39.3700 in. 1 m = 3.28 ft	1 in. = 25.4 mm 1 ft = 304.8 mm 1 in. = 2.54 cm 1 ft = 30.48 cm 1 in. = 0.0254 m 1 ft = 0.3048 m
Area	1 mm^2 = 1.55E-3 in^2 1 mm^2 = 1.0764E-5 ft^2 1 cm^2 = 0.155 in^2 1 cm^2 = 1.07E-3 ft^2 1 m^2 = 1550 in^2 1 m^2 = 10.76 ft^2	1 in^2 = 645.16 mm^2 1 ft^2 = 92,903 mm^2 1 in^2 = 6.4516 cm^2 1 ft^2 = 929.03 cm^2 1 in^2 = 6.4516E-4 m^2 1 ft^2 = 0.0929 m^2
Volume	1 mm^3 = 6.1024E-4 in^3 1 mm^3 = 3.5315E-8 ft^3 1 cm^3 = 0.061024 in^3 1 cm^3 = 3.5315E-5 ft^3 1 m^3 = 61024 in^3 1 m^3 = 35.315 ft^3	1 in^3 = 16387 mm^3 1 ft^3 = 28.317E6 mm^3 1 in^3 = 16.387 cm^3 1 ft^3 = 28317 cm^3 1 in^3 = 1.6387E-5 m^3 1 ft^3 = 0.028317 m^3

Dimension Homogenity

- Given $L = a + b + c$
- Left hand side of equation should have the same dimension as right hand side of equation
- If L represents dimension length, then a, b, and c must also have the dimension of length – this is called dimensionally homogeneous

Example: Dimension Homogenity



Example

Given $d = \frac{PL}{AE}$ where

- d is the end deflection in m,
- P is the applied load in N,
- L is the length of the bar in m,
- A is the cross-sectional area of the bar in m^2, and
- E is the modulus of elasticity,

find units for E.

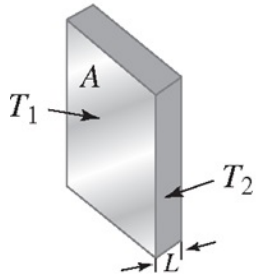
Example: Dimension Homogenity

Solution

$$d = \frac{PL}{AE} \implies m = \frac{(N)(m)}{m^2 E}$$

$$E = \frac{(N)(m)}{(m)(m^2)} = \frac{N}{m^2}$$

Example: Dimension Homogenity



Example

Given $q = kA\frac{T_1 - T_2}{L}$ where

- q is the heat transfer rate,
- k is the thermal conductivity of the solid material in $W/m \cdot^\circ C$,
- A is the area in m^2,
- $T_1 - T_2$ is the temperature difference in $^\circ C$, and
- L is the thickness of the material in m,

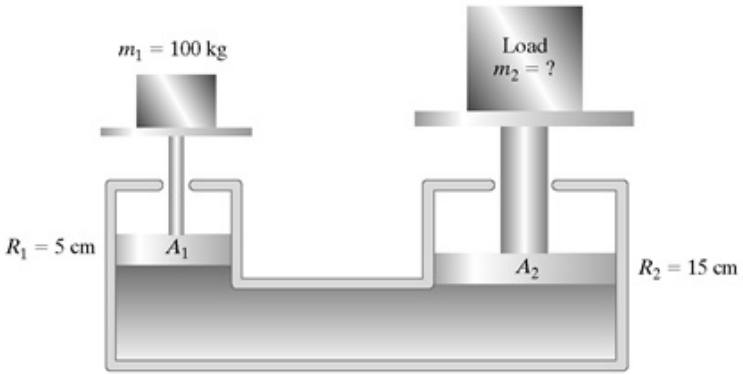
find units for q.

Example: Dimension Homogenity

Solution

$$q = kA\frac{T_1 - T_2}{L} = \left(\frac{W}{m \cdot^{\circ} C}\right)(m^2)\left(\frac{^{\circ}C}{m}\right) = W$$

Example: Numerical vs. Symbolic Solutions



Example

Given hydraulic system shown above, find m_2.

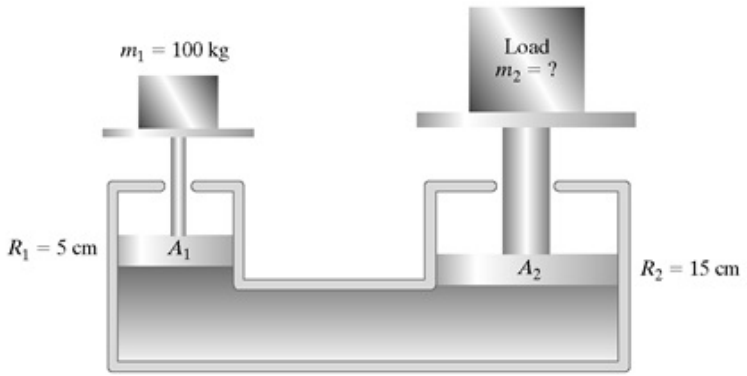
Solution (Numerical solution)

$$F_1 = m_1 g = 100kg(9.81m/s^2) = 981N$$

$$F_2 = \frac{A_2}{A_1} F_1 = \frac{\pi(0.15m)^2}{\pi(0.05m)^2}(9.81N) = 8829N$$

$$F_2 = m_2(9.81m/s^2) = 8829N \implies m_2 = 900kg$$

Example: Numerical vs. Symbolic Solutions



Solution (Symbolic solution)

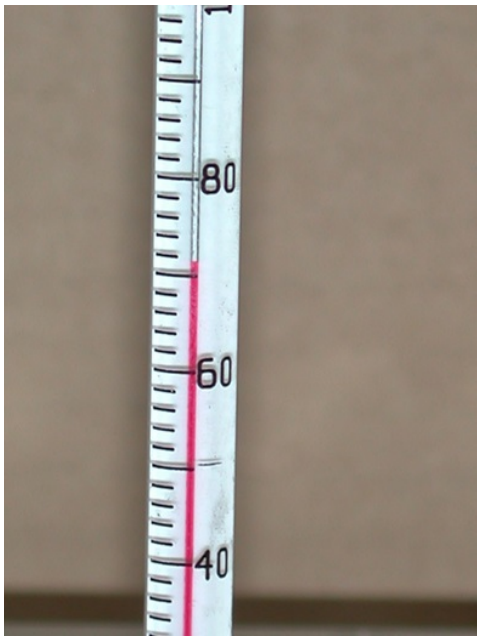
$$F_2 = \frac{A_2}{A_1} F_1 \iff m_2 g = \frac{\pi (R_2)^2}{\pi (R_1)^2} (m_1 g)$$

$$m_2 = \frac{R_2^2}{R_1^2} m_1 = \frac{(0.15m)^2}{(0.05m)^2} (100kg) = 900kg$$

Significant Digits (Figures)

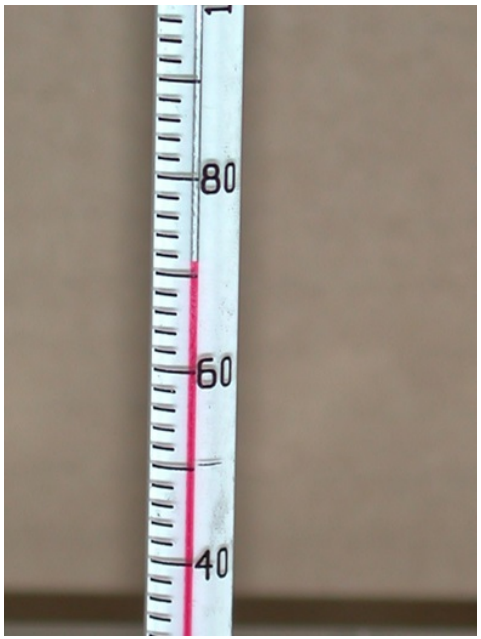
- Engineers make measurements and carry out calculations
- Engineers record the results of measurements and calculations using numbers
- Significant digits (figures) represent (convey) the extend to which recorded or computed data is dependable

Significant Digits – How to Record a Measurement



- Least count – one half of the smallest scale division
- What should we record for this temperature measurement?

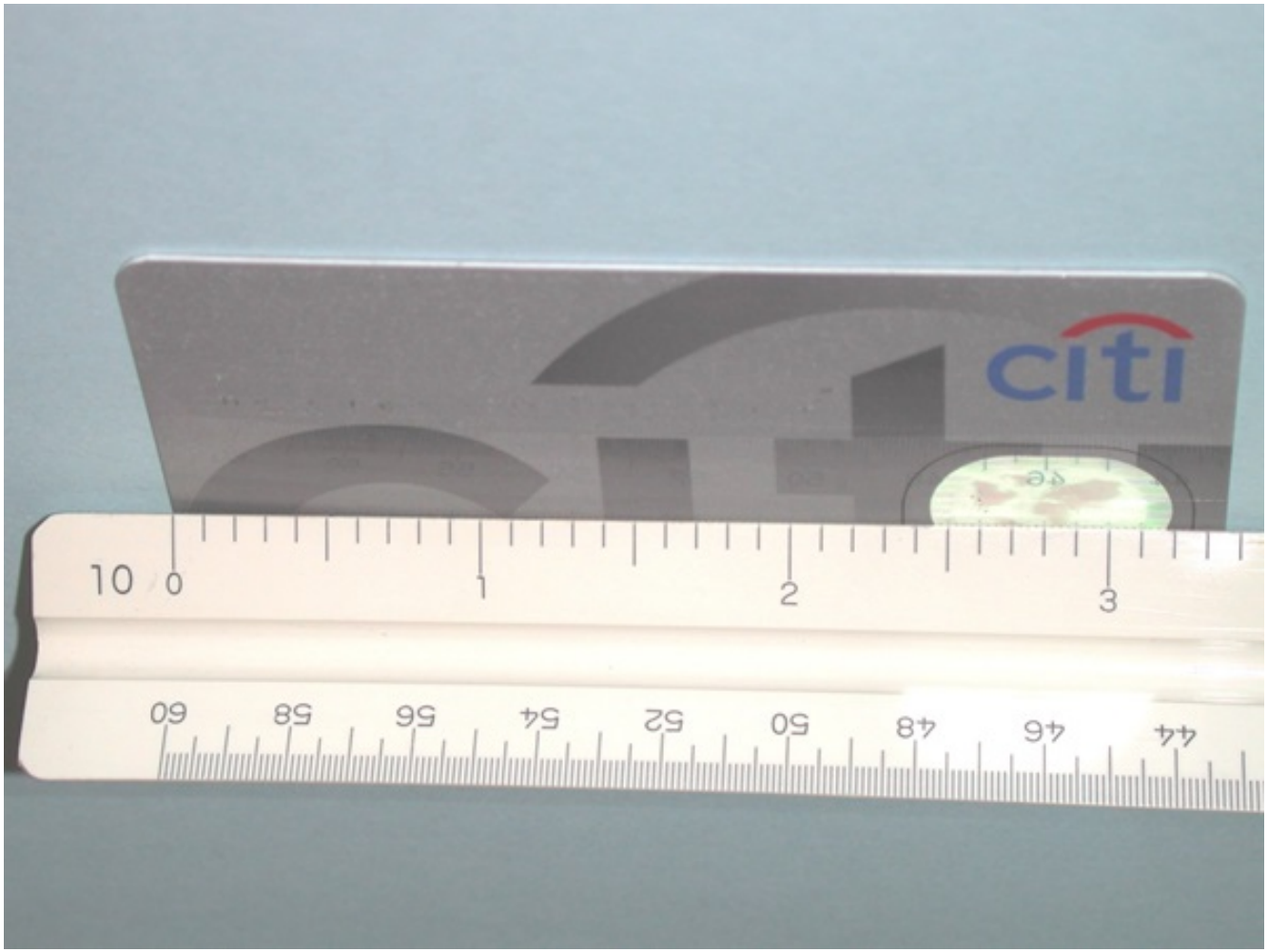
Significant Digits – How to Record a Measurement



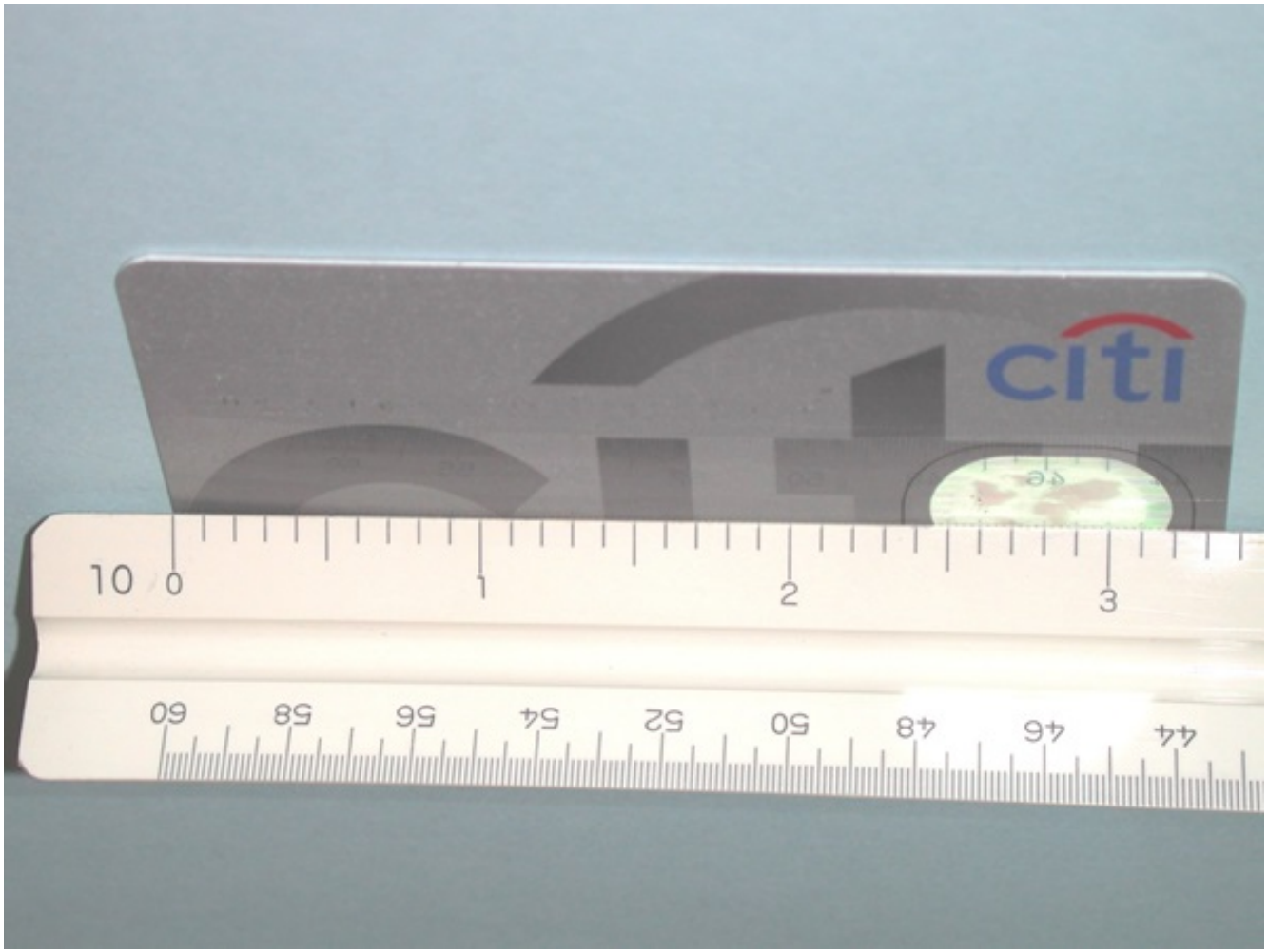
- Least count – one half of the smallest scale division
- What should we record for this temperature measurement?

$$71 \pm 1^{\circ}C$$

Significant Digits – How to Record a Measurement

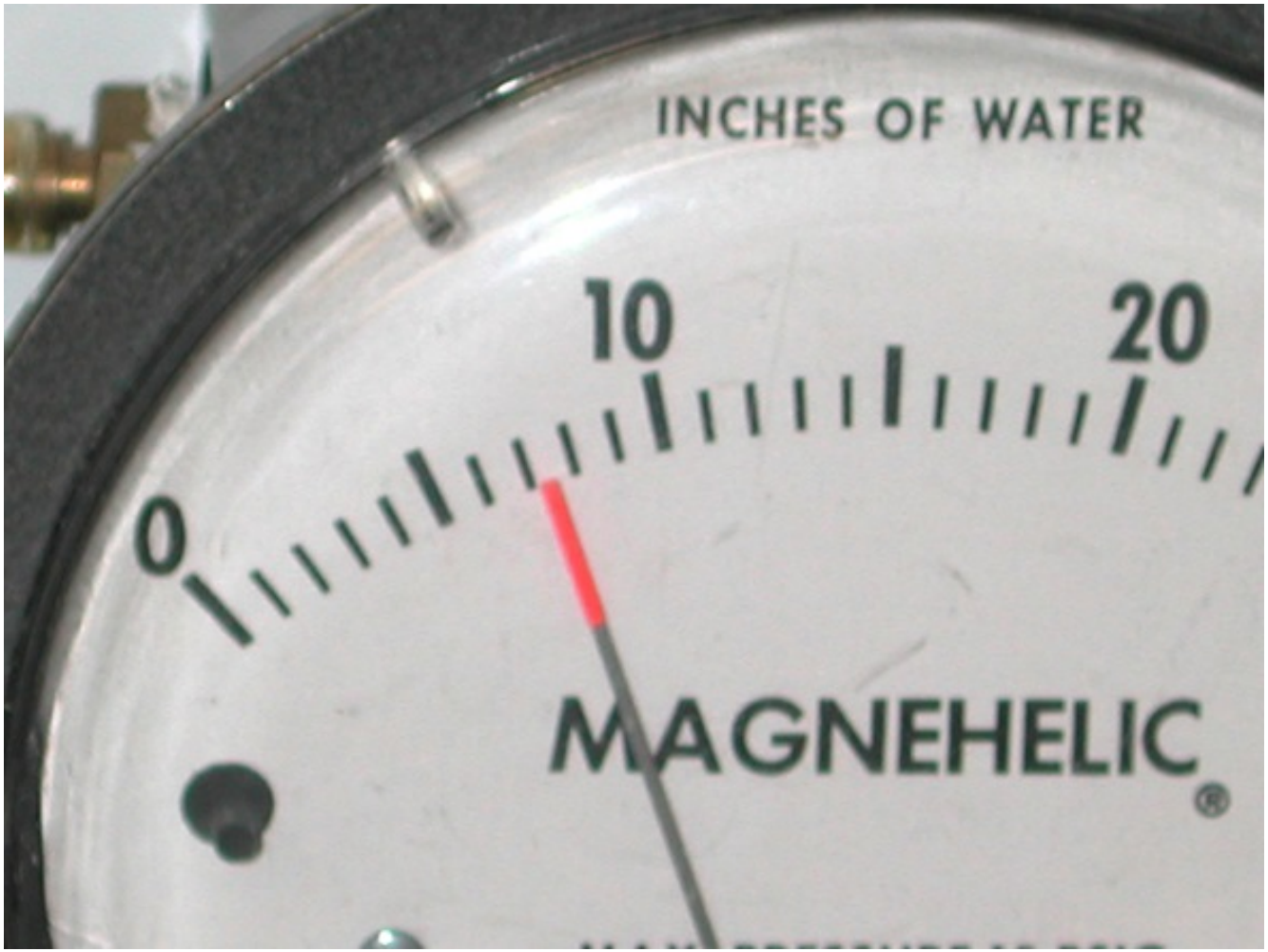


Significant Digits – How to Record a Measurement

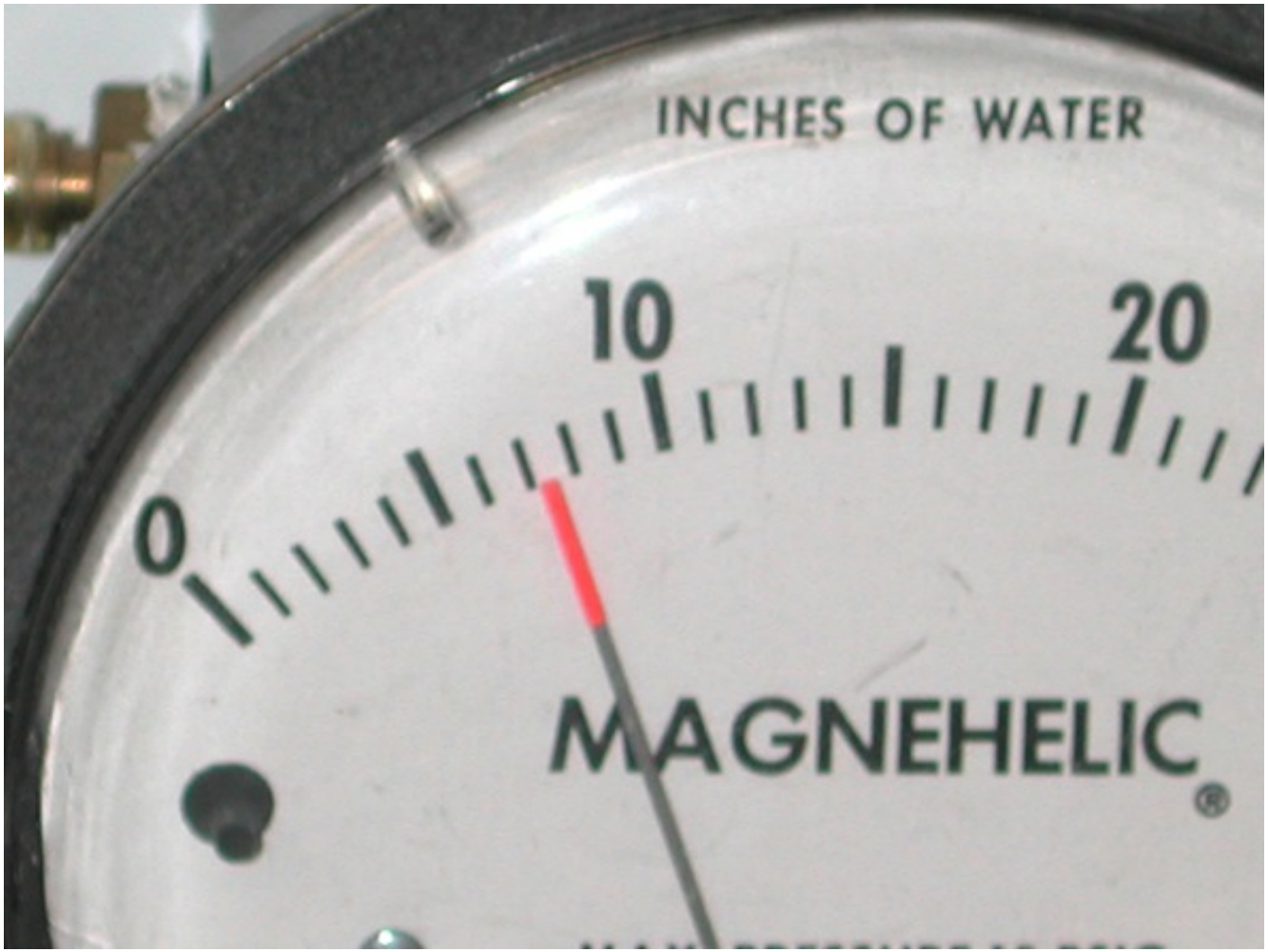


$$3.35 \pm 0.05in$$

Significant Digits – How to Record a Measurement



Significant Digits – How to Record a Measurement



$7.5 \pm 0.5in$